TN-Check: Time-Bounded CSL Model Checking of Chemical Master Equations on Tensor-Train State Vectors

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Preprint

Abstract

Time-bounded reachability on the chemical master equation (CME) is a core query in stochastic systems biology, and probabilistic model checkers answer it by uniformisation on an explicitly stored state space. The state space of a reaction network grows as the product of the copy-number ranges of its species, and explicit and hybrid engines store at least one probability vector over the relevant states. TN-Check evaluates the time-bounded until operator of CSL directly on tensor-train (TT) representations of the probability vector and the CME generator. It applies the absorbing projection of the until operator to the generator in matrix-product-operator form, runs uniformisation with TT rounding, and reports the Poisson truncation mass and the finite-state-projection (FSP) leakage alongside the result. On a shared suite exported with identical FSP semantics to Storm 1.14 and PRISM 4.10, TN-Check agrees with Storm to within $3 \cdot 10^{-9}$ on all cascade models up to $3.4 \cdot 10^{7}$ states and on toggle switches up to $1.6 \cdot 10^{5}$ states. On an n-stage gene-expression cascade with 32 copy numbers per species it answers the query for $n = 6$ to $n = 10$ ($1.1 \cdot 10^{9}$ to $1.1 \cdot 10^{15}$ states) in 3 to 18 minutes on two CPU cores and 4 GiB of memory, with every answer inside the 99% confidence interval of 10^{6} independent SSA runs. With 16 GiB, Storm does not finish $n = 6$ within an hour and PRISM runs out of memory or time on $n = 6$. The TT-CME solver of Ion et al. is faster on the largest cascades and agrees with TN-Check to within $5 \cdot 10^{-6}$ there; where an exact reference exists, TN-Check's error is one to five orders of magnitude smaller than TT-CME's.

1 Introduction

A stochastic reaction network with N species, each truncated to d copy numbers, has a finite-state-projection (FSP) state space of d^N states [12]. Continuous stochastic logic (CSL) model checking of such a model asks for probabilities such as $\Pr[\mathit{true}\ \mathrm{U}^{\leq T}\ \phi]$, the probability that a target set is reached by time T. Storm [7] and PRISM [10] compute this by uniformisation [1, 5] on an explicit sparse matrix or on a hybrid representation that still stores probability vectors explicitly. The memory cost of these vectors, $8d^N$ bytes each, bounds the models they can check.

Tensor trains (TT, also called matrix product states) [15] store a d^N vector with $O(Nd\chi^2)$ numbers, where χ is the TT rank. Earlier work showed that CME solutions of many networks have low TT rank and solved the CME directly in TT or quantised-TT format [9, 4, 8]. The inspected contributions address transient CME solutions and parameter inference. The candidate distinction here is the CSL until projection and query interface; TT compression and scalable CME evolution are established prior work, and a universal absence of earlier TT property checking has not been established.

Claim. Time-bounded CSL until on FSP-truncated CMEs can be evaluated in TT format with accuracy comparable to an explicit model checker on models that both can handle, and on models whose state space is several orders of magnitude beyond what explicit and hybrid engines handle in the same memory.

Contributions.

- An algorithm for $\Pr[\phi_1 \,\mathrm{U}^{\le T}\, \phi_2]$ on TT state vectors: the absorbing projection is applied to the generator as a column mask in MPO form, uniformisation is accumulated on scalars, and the result carries the Poisson tail and the FSP leakage as explicit error terms (Section 3).
- A PRISM exporter with the same FSP sink semantics as the TT compiler, which makes TN-Check, Storm and PRISM answer the identical query on the identical CTMC (Section 4).
- An evaluation against Storm (sparse and hybrid engines), PRISM (explicit, sparse, hybrid, MTBDD), the TT-CME solver of Ion et al. [8], an independent scipy reference, and Gillespie simulation [6], on models from 50 to $1.1 \cdot 10^{15}$ states (Section 5).

2 Background and related work

CSL model checking of CTMCs. For a CTMC with generator Q (column convention, $\dot{p} = Qp$), $\Pr[\phi_1 \,\mathrm{U}^{\le T}\, \phi_2]$ equals the probability mass in $\mathrm{Sat}(\phi_2)$ at time T of the chain in which every state outside $\mathrm{Sat}(\phi_1) \setminus \mathrm{Sat}(\phi_2)$ is absorbing [1]. Uniformisation writes $e^{Q'T}p_0 = \sum_k \mathrm{Pois}(k; qT)\, P^k p_0$ with $P = I + Q'/q$ and q at least the largest exit rate; the Poisson series is truncated with Fox–Glynn style bounds [5]. Storm and PRISM implement this on sparse matrices and on MTBDD-based hybrid representations. STAMINA [13] truncates infinite CTMCs on the fly and bounds the probability of the truncated part; Mikeev et al. [11] check timed-automaton properties on large Markov population models with on-the-fly state-space truncation. These methods keep an explicit vector over the retained states.

Tensor methods for the CME. Kazeev et al. [9] solved the CME in quantised TT format; Dolgov and Khoromskij [4] solved it with a simultaneous space-time TT representation and the AMEn solver [3]; Ion et al. [8] use TT for the CME and for parameter inference and provide the TT-CME code we compare against. Buchholz et al. [2] study hierarchical-Tucker compression of solution vectors in Kronecker-structured Markov models. Nicholson and Gingrich [14] use tensor networks for rare-event probabilities in reaction-diffusion CMEs.

3 Approach

Generator. Every reaction r with stoichiometry v_r and a propensity that factors over species, $a_r(n) = \prod_k f_{r,k}(n_k)$ (mass action, Hill and Michaelis–Menten kinetics do), contributes a rank-one MPO $\bigotimes_k S_{r,k}F_{r,k}$, where $F_{r,k} = \mathrm{diag}(f_{r,k})$ and $S_{r,k}$ shifts species k by $v_{r,k}$, plus a rank-one diagonal term $-\bigotimes_k F_{r,k}$. Shifts that leave the box are dropped from the off-diagonal part but kept on the diagonal, so probability that leaves the box is lost; this is the FSP sink. The sum is rounded to a compact MPO.

Absorbing projection. Let $m = \mathrm{Sat}(\phi_1) \odot (1 - \mathrm{Sat}(\phi_2))$. For axis-aligned atomic propositions both satisfaction sets are rank-one TTs, so m has rank one and $Q' = Q\,\mathrm{diag}(m)$ has the MPO rank of Q. Zeroing columns makes the states outside m absorbing while keeping the inflow into $\mathrm{Sat}(\phi_2)$.

Uniformisation in TT format. With $q \geq \max_n \sum_r a_r(n)$, computed from the per-species factors as $\sum_r \prod_k \max f_{r,k}$, and $P = I + Q'/q$, TN-Check iterates $x_{k+1} = \mathcal{R}_{\chi,\varepsilon}(Px_k)$, where $\mathcal{R}$ is TT-SVD rounding [15] to rank at most χ and relative accuracy ε. Only scalars are accumulated:

$$\hat{p} = \sum_{k=0}^{K} w_k \,\langle \mathrm{Sat}(\phi_2), x_k \rangle, \qquad \hat{m} = \sum_{k=0}^{K} w_k \,\langle \mathbf{1}, x_k \rangle, \qquad w_k = \mathrm{Pois}(k; qT),$$

with K the smallest index whose Poisson right tail is below 10^{-10}. Inner products with rank-one masks cost $O(Nd\chi^2)$, so no full vector is formed.

Error terms. TN-Check reports three quantities with every answer. The Poisson tail $\tau = 1 - \sum_k w_k$ bounds the uniformisation truncation, because every exact iterate has $\|P^k p_0\|_1 \leq 1$. For exact iterates, the deficit $s = 1 - \hat{m} - \tau$ bounds the FSP error from above: the probability of the untruncated CME lies in $[\hat{p}, \hat{p} + s]$ up to the other terms [12]. The TT rounding error is estimated by rerunning with 2χ and $\varepsilon/100$ and taking the difference of the two answers; the checker widens its interval by this amount and returns INDETERMINATE when the interval contains the threshold. The rounding estimate is empirical; the Poisson and FSP terms hold for the exact iterates. For an untruncated nonexplosive CME, the latter requires the same until property and consistent initial distribution; it concerns paths that exit the box before termination. Finite-box agreement alone does not validate this infinite-state approximation. Rounding the generator and the uniformisation MPO also introduces perturbations. The reported interval is therefore a numerical diagnostic. A trace consistency check validates the recorded scalar arithmetic and threshold comparison.

Unbounded until. For unbounded until TN-Check doubles the horizon T and uses $\Pr[\mathrm{U}^{\leq T}] \leq \Pr[\mathrm{U}] \leq \Pr[\mathrm{U}^{\leq T}] + \langle m, x(T) \rangle + s$, where $\langle m, x(T) \rangle$ is the mass still in transient states, until the interval is narrower than a tolerance.

4 Implementation

TN-Check is a Python package (numpy, scipy) with a command-line interface that reads YAML, JSON or SBML networks and CSL formulas. The TT engine is `tn_check/checker/tt_transient.py`. Three components exist only for validation and share no numerical code with the engine: an explicit FSP generator assembled state by state from the propensity functions with scipy `expm_multiply` for the until probability (`reference/explicit_cme.py`); a vectorised Gillespie simulator with Wilson and Clopper–Pearson intervals [16] (`reference/ssa.py`); and a PRISM exporter (`export/prism.py`). The exporter writes one bounded integer variable per species, guards every command by the box, and redirects out-of-box firings to an absorbing sink in which all counts are reset, so the exported CTMC is the FSP chain used by the TT compiler; atomic propositions exclude the sink.

5 Evaluation

Questions. RQ1: Does TN-Check compute the same probabilities as Storm and PRISM? RQ2: How do time and memory compare with explicit, hybrid and TT baselines? RQ3: Does TN-Check

give correct answers on models that the explicit tools cannot check?

Models and properties. The suite contains the four example networks of the repository (birth–death, two-stage gene expression, a toggle switch with Hill repression, and Michaelis–Menten enzyme kinetics), the toggle switch with boxes of 200^2 and 400^2, and an n-stage cascade for $n = 2, \ldots, 10$ in which S_1 is produced at rate 1, S_i is produced with Hill propensity $x_{i-1}^2/(25 + x_{i-1}^2)$, and each species degrades at rate 0.1, with copy numbers $0, \ldots, 31$ per species. Each model has one property $\Pr[\mathit{true}\ \mathrm{U}^{\le T}\ x_k \ge c]$; for the cascade it is $x_n \ge 3$ with $T = 10n$.

Setup. All tools run in separate Modal containers: Storm 1.14.0 (Docker image `movesrwth/storm:1.14.0`) and PRISM 4.10.1 with 1 CPU and 16 GiB (Java heap 12 GiB), TN-Check and the SSA with 2 CPUs and 4 GiB, TT-CME (commit `4aebc15`, torchTT 0.5.0, Crank–Nicolson time stepping, AMEn tolerance 10^{-8}) with 2 CPUs and 8 GiB. TT-CME uses a reflecting copy-number boundary, while TN-Check, Storm and PRISM use an FSP sink; boundary excursions can therefore affect their probability difference. Time is wall-clock time of the process, memory is its peak resident set. The timeout is 600 s for the small models and 1200 s for $n = 5, 6$. The reference value is the scipy explicit computation where it fits (all models up to 10^6 states) and Storm sparse otherwise. TN-Check runs with $(\chi, \varepsilon) = (32, 10^{-8})$ and $(64, 10^{-10})$.

RQ1: agreement. Table 1 lists the absolute difference from the reference. With $\chi = 64$, TN-Check is within $7 \cdot 10^{-11}$ of the reference on birth–death and the three toggle switches, within $5.4 \cdot 10^{-9}$ on gene expression, and within $3 \cdot 10^{-9}$ on the cascades up to $n = 5$. Storm sparse and hybrid agree with the scipy reference to within $4 \cdot 10^{-11}$; PRISM agrees to within $6 \cdot 10^{-8}$ with its default settings, except on gene expression, where its sparse, hybrid and MTBDD engines return a value $7 \cdot 10^{-5}$ below the reference. The enzyme kinetics model has two conservation laws, and in the species ordering used its distribution needs TT rank 120: at $\chi = 32$ and $\chi = 64$ TN-Check is off by 0.25 and 10^{-2}, the rank-doubling estimate reports a difference of 0.24, and the checker returns INDETERMINATE for thresholds inside the widened interval; at $\chi = 128$ the error is $2 \cdot 10^{-9}$ and at $\chi = 256$ it is $6 \cdot 10^{-12}$. Across the 51 models of the repository's model library, the MPO generator matches propensities evaluated directly at 200 random states of every model to within $1.6 \cdot 10^{-13}$, and on the 47 models with at most $1.1 \cdot 10^6$ states the TT answer to a reachability property differs from the scipy reference by at most $4.1 \cdot 10^{-9}$.

RQ2: cost. Where the state space fits, Storm sparse is the fastest tool by one to two orders of magnitude: it builds only the states relevant to the property (for the cascade, the states with $x_n \le 3$) and needs 0.1 to 46 s on all models up to $n = 5$. TN-Check needs 2 to 140 s on the same models. Its cost grows with the number of uniformisation steps, which is about qT (116 to 10 665 MPO applications in this suite), and with χ^3. TT-CME is slower than TN-Check at $\chi = 64$ on all models up to $n = 4$ and faster on the cascades from $n = 5$ on (Tables 1 and 2): its implicit time stepping does not depend on q, while TN-Check's step count grows with n through both q and $T = 10n$. At AMEn tolerance 10^{-8} the TT-CME error against the exact reference is $9 \cdot 10^{-8}$ to $1.7 \cdot 10^{-5}$ across the suite, compared with $2 \cdot 10^{-11}$ to $5.4 \cdot 10^{-9}$ for TN-Check at $\chi = 64$ (excluding the enzyme model). These observed accuracy differences are not a matched-accuracy ranking: the solvers use different boundaries, numerical schemes and stopping criteria. A matched-boundary, matched-error cost comparison remains open.

model	box states	Storm sparse s	Storm sparse GB	PRISM sparse s	tn_check (χ=64) s	tn_check (χ=64) abs. err	TT-CME (ε=10^{-8}) s	TT-CME (ε=10^{-8}) abs. err
birth_death	$5.0 \cdot 10^1$	0.2	0.2	1.4	2.1	4e-11	10.1	3e-6
gene_expression	$8.0 \cdot 10^3$	0.3	0.2	2.0	9.3	5e-9	311	9e-8
toggle_switch	$1.0 \cdot 10^4$	0.2	0.2	1.6	2.8	2e-11	42.3	2e-6
toggle_d200	$4.0 \cdot 10^4$	0.4	0.2	2.4	42.2	4e-11	234	5e-7
toggle_d400	$1.6 \cdot 10^5$	0.8	0.2	9.2	140	6e-11	1763	6e-7
enzyme_kinetics	$6.4 \cdot 10^5$	0.1	0.2	1.4	26.1	1e-2	47.6	2e-5
cascade_n2	$1.0 \cdot 10^3$	0.1	0.2	1.8	2.4	5e-11	8.6	1e-6
cascade_n3	$3.3 \cdot 10^4$	0.1	0.2	1.6	2.8	1e-9	17.3	7e-7
cascade_n4	$1.0 \cdot 10^6$	1.2	0.2	12.9	22.0	2e-9	42.2	3e-7
cascade_n5	$3.4 \cdot 10^7$	45.7	2.5	349	102	3e-9	66.9	2e-7
cascade_n6	$1.1 \cdot 10^9$	TO	–	TO	571	–	153	–

Table 1: Wall time (s), peak memory (GB) and absolute error against the reference. TO: timeout. For $n = 6$ the TT error column is empty because no exact reference exists; see Table 2.

n	box states	tn_check χ=32	tn_check χ=64	$\lvert\Delta\rvert$	s (χ=32)	s (χ=64)	TT-CME ε=10^{-8}	SSA 99% CI (10^6 runs)
3	$3.3 \cdot 10^4$	0.886737	0.886737	4e-08	1.5	2.8	0.886738 (17.3)	[0.8859, 0.8875]
4	$1.0 \cdot 10^6$	0.809520	0.809520	1e-07	12.3	22.0	0.809520 (42.2)	[0.8088, 0.8108]
5	$3.4 \cdot 10^7$	0.727022	0.727022	3e-07	59.2	102	0.727023 (66.9)	[0.7255, 0.7278]
6	$1.1 \cdot 10^9$	0.644761	0.644766	4e-06	178	571	0.644766 (153)	[0.6434, 0.6459]
7	$3.4 \cdot 10^{10}$	0.566100	0.566107	7e-06	288	860	0.566107 (210)	[0.5652, 0.5677]
8	$1.1 \cdot 10^{12}$	0.492950	0.492963	1e-05	342	1366	0.492962 (379)	[0.4921, 0.4946]
9	$3.5 \cdot 10^{13}$	0.426288	0.426299	1e-05	1019	2633	0.426297 (648)	[0.4243, 0.4268]
10	$1.1 \cdot 10^{15}$	0.366437	0.366456	2e-05	1077	2976	0.366451 (790)	[0.3660, 0.3685]

Table 2: Cascade models with 32 copy numbers per species. $|\Delta|$: change between the $\chi = 32$ and $\chi = 64$ runs. Times in seconds.

RQ3: beyond explicit engines. For $n = 6$ the relevant state space explored by Storm has $1.3 \cdot 10^8$ states and $1.2 \cdot 10^9$ transitions. Storm sparse builds the model in 568 to 906 s and does not finish the transient analysis within 3 500 s; its hybrid engine stops with a full BDD unique table; PRISM explicit runs out of its 12 GiB Java heap, and PRISM sparse and hybrid do not finish within 1 200 s. For $n = 7$ Storm sparse is still building the model after 2 400 s; the relevant state space has about $4 \cdot 32^6 \approx 4.3 \cdot 10^9$ states, so a single probability vector over it takes 34 GB. TN-Check answers $n = 6, \ldots, 10$ with $\chi = 32$ in 3 to 18 minutes and with $\chi = 64$ in 10 to 50 minutes, using at most 0.7 GB of memory (Table 2). Each answer lies inside the Wilson 99% interval of 10^6 SSA runs; the change from $\chi = 32$ to $\chi = 64$ is at most $1.9 \cdot 10^{-5}$; and the $\chi = 64$ answers agree with TT-CME at tolerance 10^{-8} to within $5 \cdot 10^{-6}$.

Small-model rank stress test. Six model/horizon cases comprise a birth–death reachability control, an empty-goal leaking control, a 10×10 toggle switch at $T = 1, 5$, and a 6^3 three-stage cascade at $T = 1, 5$. At each requested rank $1, 2, 4, 8$, the checker reports the solution computed at twice that rank and widens its interval using the rank difference. An independently enumerated generator is solved by both sparse and dense matrix exponentials. The maximum difference between these references is $1.12 \cdot 10^{-15}$; all 24 reported intervals contain the finite-box reference within a 10^{-9} comparison tolerance. The largest probability error at requested rank 1 is 0.1116, while at requested rank 4 or 8 it is below $2.8 \cdot 10^{-11}$. These finite tests measure numerical behavior across rank caps; the rounding term remains an empirical estimate. Raw outputs are in

`results/adversarial_before.jsonl`.

Finite-box residual witness. A fresh frozen diagnostic enumerates an independent substochastic uniformisation matrix P over the box and records every approximate TT iterate x_k. In exact arithmetic, with $x_0 = p_0$, define $d_j = \|x_j - Px_{j-1}\|_1$. Contractivity gives $\|x_k - P^k p_0\|_1 \le \sum_{j=1}^{k} d_j$ by induction, hence the goal-scalar error is at most $B = \tau + \sum_{k=0}^{K} w_k \sum_{j=1}^{k} d_j$. This residual includes rounded-operator perturbations because P is independently enumerated. Twenty runs use boxes of 10, 100 and 216 states, horizons 1 and 5, and ranks 1, 2, 4, 8. All computed envelopes contain the sparse-exponential reference within 10^{-12}; low-rank toggle radii range from 1.03 to 3.30 and therefore yield no useful probability restriction. A deliberately injected identity operator gives zero reachability at both ranks 1 and 2, while the reference is 0.075866; rank agreement misses this fault and the residual radius is approximately 2. This is an artificial control, not a failure observed in the unmodified engine. Across all 22 runs, TT iteration with witness evaluation takes 0.177 s, witness evaluation 0.029 s, and explicit transient solves 0.046 s (one CPU thread; indicative microbenchmarks). The witness enumerates the box and sacrifices TT memory scaling. Floating-point residuals and Poisson weights are not outward-rounded, so this implementation remains a diagnostic rather than a certified enclosure. Code and every row are in `benchmarks/scripts/residual_witness.py` and `results/residual_witness.jsonl`.

6 Limitations

The rounding term of the error report is an empirical estimate; a small change between ranks does not establish a universal error bound. The rounded MPO perturbation is outside the Poisson and FSP bounds for exact iterates, and the evaluation covers one scalable family (a feed-forward cascade) beyond the reach of explicit tools; networks with feedback across many species can have higher TT ranks. Models whose reachable set is much smaller than the copy-number box, such as networks with conservation laws, are better served by an explicit engine that explores reachable states. The uniformisation step count grows with the maximal exit rate over the box, so stiff networks with large copy-number bounds are expensive. TN-Check handles atomic propositions on single species and their Boolean combinations; nested probability operators are not part of this evaluation.

7 Conclusion

Applying the until projection to the CME generator in MPO form and running uniformisation on TT vectors lets a CSL query that explicit engines answer exactly on 10^7 states be answered with errors of order 10^{-9} on those models and within Monte Carlo confidence intervals on 10^9 to 10^{15} states. All raw outputs, job definitions and scripts are in the repository.

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